

Compact formulation of the transparent-absorbing isotropic interface electromagnetic problem

SALVADOR BOSCH,^{1,*} JOSEP FERRÉ-BORRULL,² JORDI SANCHO-PARRAMON,³ AND ORIOL ARTEAGA¹

¹University of Barcelona, Dep. Física Aplicada, Martí i Franqués 1, 08028 Barcelona, Spain

²Universitat Rovira i Virgili, Eng. Electrònica, Elèctrica i Automàtica, Campus Sescelades, Avinguda Països Catalans, 26, 43007 Tarragona, Spain

³Ruder Boskovic Institute, Laboratory for Optics and Optical Thin Films, Bijenicka cesta 54, 10000 Zagreb, Croatia

*Corresponding author: sbosch@ub.edu

Received 13 June 2022; revised 28 July 2022; accepted 19 August 2022; posted 19 August 2022; published 6 September 2022

A complete formulation of the electromagnetic problem corresponding to the light incidence from a transparent to an absorbing medium (isotropic materials) is developed. According to the standard separation in s and p polarization cases, we explicitly obtain all the relevant formulas that relate the polarization and Poynting vectors of the reflected and transmitted beams with the incident ones. Overall, the procedure is compact since it is short and complete. © 2022 Optica Publishing Group

<https://doi.org/10.1364/JOSAA.466318>

1. INTRODUCTION

One of the most relevant historical results in the study of light-matter interaction was the success of the application of the electromagnetic theory to the reflection and refraction at a plane boundary. Once the law explaining the bending of rays at the interface was known, the big question remaining was to find the “ratio” between the reflected (or refracted) ray and the incident one. Before the development of the electromagnetic theory of light, Fresnel found the right formulas by modeling light beams as transverse elastic waves. Later, within the electromagnetic scheme, light consists of propagating electric and magnetic fields, conveying energies that may be calculated by means of the Poynting vector. It is customary in optics to focus specifically on the behavior of the electric field of the waves, which leads to the splitting of the problem into s and p polarization cases. For the case of transparent materials, the Fresnel formulas summarize the achievements [1–3].

In short, these formulas define the coefficients that represent the ratios between the moduli of the reflected and incident (or transmitted and incident) electric field vectors. Thus, for the s polarization case, these vectors are collinear, while for the p polarization case, they are not. A real positive (negative) value of the coefficient indicates that the presumed directions of the electric field vectors are correct (opposite). Therefore, the interpretation of the results is not so simple for the p case, as the incidence angle changes from 0° to 90° . Complex values for the coefficients require additional explanations, and we hope the present work also contributes to the clear interpretation of this topic. The relevance of the above ideas in electromagnetic optics

is demonstrated by the detailed explanations, figures, etc., that current prominent optics textbooks always include [4,5].

Restricting ourselves to the isotropic case, the generalization to absorbing materials is also of major practical importance. The most general problem can be fully covered in the framework of the electromagnetic theory of light [6–14], with correspondingly higher mathematical complexity, but, in the particular case of a transparent-absorbing interface, simpler formulations can be developed [15–19]. Furthermore, refraction and propagation of light in absorbing media are of current interest in many applications of photonics and optics [20].

Following the developments in [21,22], the aim of the present work is to present a compact formulation of the transparent-absorbing isotropic interface problem, i.e., when the first medium is transparent and the second one is (in general) absorbing. Due to the symmetry of the plane interface, it is evident that any incident field vibrating perpendicularly to the incidence plane will give rise to reflected and transmitted fields also vibrating in the incidence plane, i.e., to collinear vectors. Since the interpretation of a coefficient affecting collinear vectors is simple independently of the angle of incidence, the key point in our procedures consists of, first, working with the field perpendicular to the incidence plane (electric for s polarization and magnetic for the p case); later, we will write the components of the field lying in the incidence plane in terms of the previous field. Thus, for s polarization, our formulas will coincide with the standard Fresnel formulas, but for p polarization, we will first compute the magnetic field and, subsequently, the electric field by using the consistent formulas.

In summary, taking advantage of the compact vectorial formulation that we present here, it is possible to obtain the complete set of useful results with little computational effort. Our systematic procedure leads to some new formulas that generalize the Fresnel equations for the transparent case and allow a complete understanding for the polarization issues in the general case, including the study of the Poynting vector in absorbing material.

The results of this work can be important when using very thin layers of absorbing materials, as, for example, in current solar cell applications.

2. PRELIMINARIES

A. Inhomogeneous Plane Waves

Monochromatic inhomogeneous plane waves are the simplest solutions of Maxwell equations in absorbing materials. This is immediately proved by checking that

$$\vec{E}(\vec{r}) = \vec{E}_0 \exp(i\vec{k} \cdot \vec{r}), \quad \vec{H}(\vec{r}) = \vec{H}_0 \exp(i\vec{k} \cdot \vec{r}), \quad (1)$$

with $\vec{k} = \text{Re}(\vec{k}) + i\text{Im}(\vec{k})$, fulfill the wave equation in an isotropic medium with complex permittivity $\varepsilon = \varepsilon_d + i\frac{\sigma}{\varepsilon_0\omega}$, provided that

$$\vec{k} \cdot \vec{E}_0 = 0, \quad \vec{k} \cdot \vec{H}_0 = 0, \quad (2)$$

$$\vec{H}_0 = \frac{1}{\omega\mu_0\mu} \vec{k} \times \vec{E}_0 = K^H \vec{k} \times \vec{E}_0, \quad (3)$$

$$\vec{E}_0 = \frac{-1}{\omega\varepsilon_0\varepsilon} \vec{k} \times \vec{H}_0 = K^E \vec{k} \times \vec{H}_0, \quad (4)$$

$$(\vec{k})^2 = \vec{k} \cdot \vec{k} = (k^x)^2 + (k^y)^2 + (k^z)^2 = \varepsilon \frac{\omega^2}{c^2} = \varepsilon k_0^2 = CT (\in \mathbb{C}). \quad (5)$$

Equations (2)–(4) relate the complex amplitudes of the waves and the complex wave vector, and Eq. (5) is the dispersion equation. Note that this dispersion equation does not contain the quantity $|\vec{k}|^2 = \vec{k} \cdot \vec{k}^* (\in \mathbb{R})$ but $(\vec{k})^2 = \vec{k} \cdot \vec{k} = \varepsilon k_0^2 (\in \mathbb{C})$ instead. In the present work, we will focus on non-magnetic materials ($\mu = 1$). The meaning of an inhomogeneous wave becomes evident by expanding the complex amplitude in the form

$$\vec{E}(\vec{r}) = \vec{E}_0 \exp(-\text{Im}(\vec{k}) \cdot \vec{r}) \exp(i\text{Re}(\vec{k}) \cdot \vec{r}), \quad (6)$$

showing that the planes of constant phase and those of constant amplitude are defined, respectively, by the real and imaginary parts of the complex wave vector $\vec{k}$.

Due to the monochromatic assumption, the calculation of the complex amplitudes of all the waves coexisting in the interface is enough to directly know the instantaneous values of all the fields by multiplying the relevant complex amplitude with the time phase factor $\exp(-i2\pi t/T)$ and taking the real part afterwards.

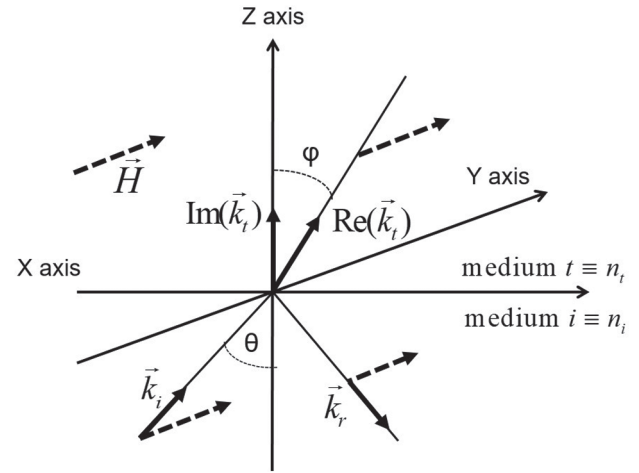


Fig. 1. Coordinate axes used in our developments. The $\vec{H}$ fields are represented, as corresponding to the p polarization case.

B. Coordinate System

Figure 1 illustrates some vectors and the coordinate axes used in our work. Here the plane of incidence is the $(x-z)$ plane, and the magnetic fields $\vec{H}$ shown are supposed to be perpendicular to the incidence plane, as corresponding to the p polarization case (for s polarization, the perpendicular field would be simply $\vec{E}$). Dealing with isotropic materials, the symmetry of the configuration guarantees that the incident, reflected, and transmitted $\vec{H}$ fields vibrate perpendicular to the incidence plane, i.e., in our y direction. Moreover, the transmitted wave will be inhomogeneous, and the planes of constant amplitude in the absorbing (t) medium are defined by the imaginary part of the complex wave vector $\text{Im}(\vec{k}_t)$, which must be pointing in the upper z direction due to the symmetry of the configuration.

C. Light Refraction

For the two media i and t , assume that the optical constants are, respectively,

$$i: n_i^2 = \varepsilon_i, \mu_i (\text{all} \in \mathbb{R}); \quad t: (n_t + i\kappa_t)^2 = \varepsilon_t (\in \mathbb{C}), \mu_t (\in \mathbb{R}). \quad (7)$$

For an incidence angle θ in the incidence plane $(x-z)$, the two components of the wave vectors can be directly calculated by taking into account that the tangential components (here the x coordinates) are the same for the incident, reflected, and transmitted wave vectors [16]:

$$k_i^x = k_t^x = k_0 n_i \sin(\theta), \quad k_i^z = \sqrt{(k_0 n_i)^2 - (k_i^x)^2} (\in \mathbb{R}),$$

$$k_t^z = \sqrt{(k_0 (n_t + i\kappa_t))^2 - (k_i^x)^2} (\in \mathbb{C}), \quad \left(\frac{\omega}{c} = k_0\right). \quad (8)$$

Due to the isotropy of the materials and the symmetry of the configuration, the vector of $\text{Im}(\vec{k}_t)$ points in the direction of the positive z axis. Finally, the direction of the phase propagation in medium t (φ) is given by the angle {expression (6) in [18]}

$$\varphi = \tan^{-1} \left(\frac{k^x}{\text{Re}(k_t^z)} \right) = \cos^{-1} \left(\frac{\text{Re}(k_t^z)}{|\text{Re}(k_t)|} \right). \quad (9)$$

Thus, the knowledge of the optical constants and the incidence angle is enough to determine all the phase propagation characteristics of the inhomogeneous wave in the second medium. Next, we will focus on the polarization of light.

3. FORMULATION OF THE P POLARIZATION CASE

A. Computing the $\vec{E}$ and $\vec{H}$ Fields

Although it is not the usual procedure for studying the p polarization case [1–3], we will follow the process in [21,22], and we will write the components of the field contained in the incidence plane (electric field) in terms of the field perpendicular to the incidence plane (magnetic field). This procedure will be later used for the s polarization case, and we will take advantage of the whole method to illustrate several common polarization issues, including the study of the Poynting vector.

For p polarization, the $\vec{H}$ fields are perpendicular to the incidence plane. Thus, we may write the $\vec{H}$ and $\vec{E}$ fields (incident, reflected, and transmitted) in component form as follows:

$$\begin{aligned} i: & \begin{bmatrix} 0 \\ H_i^y \\ 0 \end{bmatrix}, \begin{bmatrix} E_i^x \\ 0 \\ E_i^z \end{bmatrix}; r: \begin{bmatrix} 0 \\ H_r^y \\ 0 \end{bmatrix}, \begin{bmatrix} E_r^x \\ 0 \\ E_r^z \end{bmatrix}; \\ t: & \begin{bmatrix} 0 \\ H_t^y \\ 0 \end{bmatrix}, \begin{bmatrix} E_t^x \\ 0 \\ E_t^z \end{bmatrix} \left\{ \begin{array}{l} \text{all} \\ E_{it}^{xyz}, H_{it}^{xyz}, \\ \in \mathbb{C} \end{array} \right. \end{aligned} \quad (10)$$

where all the scalar quantities therein may be complex numbers. For the wave vectors, we have

$$\vec{k}_i = \begin{bmatrix} k_i^x \\ 0 \\ k_i^z \end{bmatrix}; \vec{k}_r = \begin{bmatrix} k_i^x \\ 0 \\ -k_i^z \end{bmatrix}; \vec{k}_t = \begin{bmatrix} k_i^x \\ 0 \\ k_t^z \end{bmatrix} \left\{ \begin{array}{l} k_i^x \equiv \text{same} \\ k_i^x, k_i^z \in \mathbb{R}, \\ k_t^z \in \mathbb{C} \end{array} \right. \quad (11)$$

where it is important to note that k_i^x and k_i^z are real-valued quantities while k_t^z is complex. Note also that we have made explicit the fact that all the x components of all the wave vectors are equal to the same value k_i^x .

Using the formulas in Eqs. (3) and (4), we may write

$$\begin{aligned} \vec{E}_i &= K_i^E (\vec{k}_i \times \vec{H}_i) = K_i^E \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ k_i^x & 0 & k_i^z \\ 0 & H_i^y & 0 \end{vmatrix} \\ &= K_i^E H_i^y \begin{pmatrix} -k_i^z \\ 0 \\ k_i^x \end{pmatrix}, \quad K_i^E = \frac{-1}{\omega \varepsilon_0 \varepsilon_i}, \\ \vec{E}_r &= K_i^E (\vec{k}_r \times \vec{H}_r) = K_i^E \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ k_i^x & 0 & -k_i^z \\ 0 & H_r^y & 0 \end{vmatrix} = K_i^E H_r^y \begin{pmatrix} k_i^z \\ 0 \\ k_i^x \end{pmatrix}, \\ \vec{E}_t &= K_t^E (\vec{k}_t \times \vec{H}_t) = K_t^E \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ k_i^x & 0 & k_t^z \\ 0 & H_t^y & 0 \end{vmatrix} = K_t^E H_t^y \begin{pmatrix} -k_t^z \\ 0 \\ k_i^x \end{pmatrix}, \\ K_t^E &= \frac{-1}{\omega \varepsilon_0 \varepsilon_t}. \end{aligned} \quad (12)$$

B. Continuity of the Fields at the Interface

The continuity of the tangential components of the $\vec{H}$ (y coordinate) and $\vec{E}$ (x coordinate) fields may be written as follows:

$$\begin{cases} H_i^y + H_r^y = H_t^y, \\ K_i^E H_i^y (-k_i^z) + K_i^E H_r^y (k_i^z) = K_t^E H_t^y (-k_t^z). \end{cases} \quad (13)$$

We can define the reflection and refraction coefficients for the $\vec{H}$ field, explicitly for the y component:

$$\begin{aligned} K_i^E H_i^y (-k_i^z) + K_i^E H_r^y (k_i^z) &= K_t^E H_t^y (-k_t^z) \Rightarrow \\ \Rightarrow K_i^E k_i^z (H_i^y - H_r^y) &= K_t^E k_t^z H_t^y, \\ r_p^y &\equiv \frac{H_r^y}{H_i^y}, \quad t_p^y \equiv \frac{H_t^y}{H_i^y} \quad \begin{cases} 1 + r_p^y = t_p^y \\ K_i^E k_i^z (1 - r_p^y) = K_t^E k_t^z t_p^y \end{cases} \Rightarrow \\ t_p^y &\equiv \frac{H_t^y}{H_i^y} = \frac{2 K_i^E k_i^z}{K_i^E k_i^z + K_t^E k_t^z} = \frac{2 \varepsilon_t k_i^z}{\varepsilon_t k_i^z + \varepsilon_i k_t^z}, \\ r_p^y &\equiv \frac{H_r^y}{H_i^y} = \frac{K_i^E k_i^z - K_t^E k_t^z}{K_i^E k_i^z + K_t^E k_t^z} = \frac{\varepsilon_t k_i^z - \varepsilon_i k_t^z}{\varepsilon_t k_i^z + \varepsilon_i k_t^z}. \end{aligned} \quad (14)$$

Thus, the remaining ratios between reflected and transmitted components and the incident ones are

$$\begin{aligned} r_p^x &\equiv \frac{E_r^x}{E_i^x} = \frac{K_i^E H_r^y k_i^z}{-K_i^E H_i^y k_i^z} = -r_p^y = r_p^z \equiv \frac{E_r^z}{E_i^z}, \\ t_p^x &\equiv \frac{E_t^x}{E_i^x} = \frac{-K_t^E H_t^y k_t^z}{-K_i^E H_i^y k_i^z} = \frac{K_t^E k_t^z}{K_i^E k_i^z} t_p^y = \frac{\varepsilon_i k_t^z}{\varepsilon_t k_i^z} t_p^y, \\ t_p^z &\equiv \frac{E_t^z}{E_i^z} = \frac{K_t^E H_t^y k_i^x}{K_i^E H_i^y k_i^x} = \frac{K_t^E}{K_i^E} t_p^y = \frac{\varepsilon_i}{\varepsilon_t} t_p^y, \quad \left[\frac{K_t^E}{K_i^E} = \frac{\varepsilon_i}{\varepsilon_t} \right]. \end{aligned} \quad (15)$$

Note that we have obtained formulas for the ratios of the fields according to their coordinate axes, not according to the kind of field. Thus, r_p^y , t_p^y refer to the $\vec{H}$ field and r_p^x , t_p^x , r_p^z , t_p^z to the $\vec{E}$ field. For the $\vec{E}$ field, we show explicitly the difference between the coefficients for the tangential (x) and normal (z) components of the field. Often this difference is not made clear when exposing the results for p polarization. Our procedure differs from the most common one [1–3] and follows [21,22], and we plan to show the subsequent advantages.

The components associated with reflected light have the usual forms (and we do not comment on these results for brevity), but for transmitted light, we explicitly see in the expressions in Eq. (15) that the ratios for the x and z components corresponding to the transmitted $\vec{E}$ field (the above quantities t_p^x , t_p^z) differ by the numerical factor $R_k^z \equiv k_t^z/k_i^z$. When the second medium is transparent, $R_k^z \equiv k_t^z/k_i^z$ is a real quantity. The standard scheme for defining the Fresnel transmission coefficients consists of calculating the ratio between the length of the transmitted and incident $\vec{E}$ fields (see the procedures in [1–3]). Using the formulas for t_p^x , t_p^z given in the expressions in Eq. (15), it is easy to check by elementary projections of vectors that the length of the transmitted electric field vector is related

to the length of the incident one by any one of the two relations (equivalently): $t_p^x \frac{\cos(\theta)}{\cos(\varphi)}$, $t_p^z \frac{\sin(\theta)}{\sin(\varphi)}$.

Thus, the standard ratios between electric fields ($r_{\parallel}$, $t_{\parallel}$) used for defining the Fresnel coefficients in the transparent interface case may be written from the expressions in Eq. (15) as follows:

$$\begin{aligned} r_{\parallel} &= r_p^x = r_p^z \\ t_{\parallel} &= t_p^x \frac{\cos(\theta)}{\cos(\varphi)} = t_p^z \frac{\sin(\theta)}{\sin(\varphi)}, \end{aligned} \quad (16)$$

where the reflection coefficient ($r_{\parallel}$) has been included for completeness. It is straightforward to check that these expressions fully coincide with expressions (20) on page 42 in [1] and (2.50) in [3] so that our electromagnetic formulation of the interface problem includes the Fresnel equations in the particular case of transparent materials, but generalizes them when the second medium is absorbing, as we will see.

C. Absorption of the Second Medium as the Key Factor in the Problem

We plan to demonstrate the fundamental differences generated by the presence or not of absorption in the second medium. For conciseness, in which follows, we will explicitly write and compare the numerical results for two virtual configurations: incidence at 80° from vacuum to two different second media having (i) $n_t = 1.5$, $\kappa_t = 0.0$, and (ii) $n_t = 1.5$, $\kappa_t = 1.1$. These two virtual cases will visibly illustrate the differences arising from the presence of absorption in the second medium.

First, immediately note that the angles for the phase propagation [as computed by means of the expression in Eq. (9)] in the two virtual cases just presented are different: 41.04° (for the transparent medium) and 37.18° (for the absorbing one).

Second, recall that the phases of the x and z components corresponding to the transmitted $\vec{E}$ field (the above quantities t_p^x , t_p^z) differ by the numerical factor $R_k^z \equiv k_t^z/k_t^x$. This factor is, for our two virtual cases, $R_k^z = 6.52 \in \mathbb{R}$ ($\kappa_t = 0.0$), and $R_k^z = 7.48 + 7.32i \in \mathbb{C}$ ($\kappa_t = 1.1$).

When the second medium is not transparent, the reasoning made above for defining the Fresnel $t_{\parallel}$ coefficient as the “ratio” between moduli of electric fields has not clear sense. Since $R_k^z \equiv k_t^z/k_t^x \in \mathbb{C}$, the x and z components are not in phase; soon we will demonstrate that the oscillating transmitted $\vec{E}$ field describes an ellipse in the incidence plane, and the direction of the vibrating $\vec{E}$ field is not constant in space. Moreover, the ratio between the moduli of the transmitted and incident electric fields is not constant, unlike for the previous $R_k^z \in \mathbb{R}$ case. In fact, what does it mean to compare the moduli of the two $\vec{E}$ fields when the transmitted oscillating direction is not constant? Thus, in this general absorbing case, the full information on refraction is contained in the two coefficients t_p^x , t_p^z , and it is not possible to use a single quantity $t_{\parallel}$ for that purpose.

The behavior of the transmitted $\vec{E}$ field is defined as “longelliptical” polarization and is illustrated in [16] using formulas (25)–(27) therein. The same results can be obtained immediately with our formulation by simply noting that the transmitted $\vec{E}$ field is the sum of the transmitted x and z components of the $\vec{E}$ field. Thus, the Jones vector of the mentioned ellipse is directly given by a couple of complex numbers

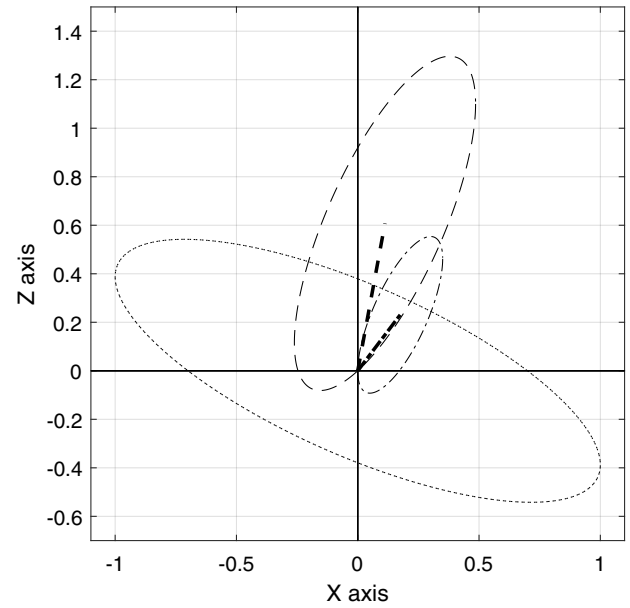


Fig. 2. Illustration of the incidence plane, showing the ellipses involved in the transmitted wave corresponding to the incidence on an absorbing substrate. Dotted line: $\vec{E}$ field (p case) or $\vec{H}$ field (s case). Dashed lines: ellipse and mean value corresponding to the Poynting vector of the p polarization case. Dotted-dashed lines: ellipse and mean value corresponding to s polarization.

(E_t^x , E_t^z), from where the angle between its major axis and the x axis (usually named ψ) and the ellipticity of the ellipse (commonly defined by χ), can be directly computed using well-known expressions for a monochromatic beam with the Jones vector (J_1 , J_2) [4] as follows:

$$\begin{aligned} \rho &\equiv \frac{J_2}{J_1} = \frac{E_t^z}{E_t^x} = \frac{k_t^x}{k_t^z} \tan(\theta), \quad \tan(2\psi) = \frac{2\text{Re}(\rho)}{1 - |\rho|^2}, \\ \sin(2\chi) &= \frac{2\text{Im}(\rho)}{1 + |\rho|^2}. \end{aligned} \quad (17)$$

Explicitly, for our $n_t = 1.5$, $\kappa_t = 1.1$ case, we have $\rho = 0.39 - 0.38i$, $\psi = -23.82^\circ$, and $|\chi| = 17.94^\circ$, ψ being the angle of the major axis with the x axis and $|\chi|$ the ellipticity. The dotted line in Fig. 2 illustrates the ellipse described by the $\vec{E}$ field in the incidence plane at any point of the interface (inside the second medium).

We have checked that the values given by these expressions, obtained directly from our vectorial products, exactly coincide with formulas (25)–(27) in [16]. Note finally that, in the other case, where k_t^z is not complex ($n_t = 1.5$, $\kappa_t = 0.0$), the ellipse becomes a line and the transmitted wave becomes a common p or transverse magnetic (TM) wave with $\rho = 0.87$, $\psi = -41.04^\circ$, $\chi = 0.0^\circ$.

D. Poynting Vector of the Transmitted Wave

Again, we do not comment on reflected light, and we concentrate only on transmission. The calculation of instantaneous values of the Poynting vector of the transmitted wave is as follows:

$$\begin{aligned}
\vec{S}_t(t) &= \text{Re} \left(\vec{E}_t \exp(-i\omega t) \right) \times \text{Re} \left(\vec{H}_t \exp(-i\omega t) \right) \\
&= \frac{1}{2} \left(\vec{E}_t \exp(-i\omega t) + \vec{E}_t^* \exp(i\omega t) \right) \\
&\quad \times \frac{1}{2} \left(\vec{H}_t \exp(-i\omega t) + \vec{H}_t^* \exp(i\omega t) \right) \\
&= \frac{1}{4} \left(\vec{E}_t \times \vec{H}_t^* + \vec{E}_t^* \times \vec{H}_t + \vec{E}_t \times \vec{H}_t \exp(-i2\omega t) \right. \\
&\quad \left. + \vec{E}_t^* \times \vec{H}_t^* \exp(i2\omega t) \right). \tag{18}
\end{aligned}$$

The mean value is $\langle \vec{S}_t(t) \rangle = \frac{1}{2} \text{Re}(\vec{E}_t \times \vec{H}_t^*)$, and the required vectorial products are

$$\begin{aligned}
\vec{E}_t \times \vec{H}_t &= K_t^E (\vec{k}_t \times \vec{H}_t) \times \vec{H}_t = K_t^E H_t^y \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -k_t^z & 0 & k_t^x \\ 0 & H_t^y & 0 \end{vmatrix} \\
&= -K_t^E (H_t^y)^2 \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix}, \\
\vec{E}_t^* \times \vec{H}_t &= -(K_t^E)^* |H_t^y|^2 \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix}^*, \\
\vec{E}_t \times \vec{H}_t^* &= -K_t^E |H_t^y|^2 \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix}, \\
\vec{E}_t^* \times \vec{H}_t^* &= -(K_t^E)^* (H_t^y)^2 \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix}^*, \tag{19}
\end{aligned}$$

from where

$$\begin{aligned}
\vec{S}_t(t) &= \frac{1}{4} \left(\vec{E}_t \times \vec{H}_t^* + \vec{E}_t^* \times \vec{H}_t + \vec{E}_t \times \vec{H}_t \exp(-i2\omega t) + \vec{E}_t^* \times \vec{H}_t^* \exp(i2\omega t) \right) \\
&= \frac{1}{2} \text{Re} \left(\vec{E}_t \times \vec{H}_t^* + \vec{E}_t \times \vec{H}_t \exp(-i2\omega t) \right) \\
&= \frac{1}{2} \text{Re} \left(-K_t^E |H_t^y|^2 \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix} - K_t^E (H_t^y)^2 \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix} \exp(-i2\omega t) \right) \left\{ \begin{array}{l} K_t^E = |K_t^E| \exp i(\phi) \\ (H_t^y)^2 = |H_t^y|^2 \exp i(2\vartheta) \end{array} \right\} \\
&= -\frac{|H_t^y|^2 |K_t^E|}{2} \text{Re} \left(\frac{K_t^E}{|K_t^E|} \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix} + \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix} \exp i(-2\omega t + \phi + 2\vartheta) \right). \tag{20}
\end{aligned}$$

The first term of the last expression corresponds to the mean value and the second term to the variations with time. Thus, the direction of the mean Poynting vector is determined (with respect to the z axis) by the angle η_p (designated here as in [17]), defined by their vector components:

$$\eta_p = \tan^{-1} \left(\text{Re} \left(\frac{K_t^E}{|K_t^E|} k_t^x \right) / \text{Re} \left(\frac{K_t^E}{|K_t^E|} k_t^z \right) \right). \tag{21}$$

The second term in the instantaneous Poynting vector describes the variations with time. The ellipse described in x and z coordinates is due to the time variations given by $\begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix} \exp i(-2\omega t + \phi + 2\vartheta)$. Thus, ϕ and ϑ are not relevant (because they affect both x and z coordinates), and the Jones vector of the ellipse described in the incidence plane is uniquely defined by the couple of complex numbers (k_t^x, k_t^z) . The dashed ellipse in Fig. 2 illustrates the variations of the Poynting vector for this p polarization. The thick line is its mean value, and the ellipse is the locus of the tip of the vector. Explicitly, for our $n_t = 1.5$, $\kappa_t = 1.1$ case, we have $\eta_p = 10.47^\circ$. Regarding the orientation and shape of the ellipse, we have $\rho = 0.39 - 0.38i$, $\psi = 66.18^\circ$, $|\chi| = 17.94^\circ$, so that the major axis is perpendicular to the major axis shown for the ellipse of the E fields (where $\psi = 23.82^\circ$). Again, we have checked that our formula gives the same values as expression (38-a) in [17]. For the case of our second transparent medium ($n_t = 1.5$, $\kappa_t = 0.0$), we have $\eta_p = 41.04^\circ$ (with respect to the z axis), also perpendicular to the vibration of the linear $\vec{E}$ field, as expected.

In summary, in the present p polarization case just developed, our compact formulation gives directly: (i) the coefficients relating the incident and transmitted fields with the incident ones (for all field components), (ii) the shapes and directions for the ellipses described by the $\vec{E}$ fields, (iii) the shapes and directions for the ellipses described by the tip of the Poynting vectors, and (iv) the direction of the mean Poynting vector. These results are described in [16,17], but our formulation is much shorter, compact, and easy to obtain. Also, we have shown that when the second medium is transparent, we re-obtain the Fresnel equations as a particular case.

4. FORMULATION OF THE S POLARIZATION CASE

The procedures are completely parallel to the previous p case. Basically, we must interchange the roles between the $\vec{E}$ and $\vec{H}$ fields when using Eqs. (3) and (4); then one has

$$\begin{aligned}
\vec{H}_i &= K_i^H (\vec{k}_i \times \vec{E}_i) = K_i^H \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ k_i^x & 0 & k_i^z \\ 0 & E_i^y & 0 \end{vmatrix} = K_i^H E_i^y \begin{pmatrix} -k_i^z \\ 0 \\ k_i^x \end{pmatrix}, \\
K_i^H &= \frac{1}{\omega \mu_0 \mu_i}, \\
\vec{H}_r &= K_i^H (\vec{k}_r \times \vec{E}_r) = K_i^H \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ k_i^x & 0 & -k_i^z \\ 0 & E_r^y & 0 \end{vmatrix} = K_i^H E_r^y \begin{pmatrix} k_i^z \\ 0 \\ k_i^x \end{pmatrix}, \\
\vec{H}_t &= K_t^H (\vec{k}_t \times \vec{E}_t) = K_t^H \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ k_t^x & 0 & k_t^z \\ 0 & E_t^y & 0 \end{vmatrix} = K_t^H E_t^y \begin{pmatrix} -k_t^z \\ 0 \\ k_t^x \end{pmatrix}, \\
K_t^H &= \frac{1}{\omega \mu_0 \mu_t},
\end{aligned} \tag{22}$$

from where

$$\begin{aligned}
r_\perp \equiv r_s^y &\equiv \frac{E_r^y}{E_i^y} = \frac{K_i^H k_i^z - K_t^H k_t^z}{K_i^H k_i^z + K_t^H k_t^z} = \frac{\mu_t k_i^z - \mu_i k_t^z}{\mu_t k_i^z + \mu_i k_t^z}, \\
t_\perp \equiv t_s^y &\equiv \frac{E_t^y}{E_i^y} = \frac{2K_i^H k_i^z}{K_i^H k_i^z + K_t^H k_t^z} = \frac{2\mu_t k_i^z}{\mu_t k_i^z + \mu_i k_t^z}, \\
r_s^x &\equiv \frac{H_r^x}{H_i^x} = \frac{K_i^H E_r^y k_i^z}{-K_i^H E_i^y k_i^z} = -r_s^y = r_s^z \equiv \frac{H_r^z}{H_i^z}, \\
\left[\frac{K_t^H}{K_i^H} = \frac{\mu_i}{\mu_t} \right], \\
t_s^x &\equiv \frac{H_t^x}{H_i^x} = \frac{-K_t^H E_t^y k_t^z}{-K_i^H E_i^y k_i^z} = \frac{\mu_i k_t^z}{\mu_t k_i^z} t_s^y, \\
t_s^z &\equiv \frac{H_t^z}{H_i^z} = \frac{K_t^H k_i^x}{K_i^H k_i^x} t_s^y = \frac{\mu_i}{\mu_t} t_s^y.
\end{aligned} \tag{23}$$

The ratios for the $\vec{H}$ field have also been obtained, but usually only the coefficients ($r_\perp$, $t_\perp$) for the $\vec{E}$ field are of interest. The explicit final formula for the transmitted Poynting vector is

$$\begin{aligned}
\vec{S}_t(t) &= \frac{|E_t^y|^2 |K_t^H|}{2} \text{Re} \left(\frac{K_t^H}{|K_t^H|} \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix} \right. \\
&\quad \left. + \begin{pmatrix} k_t^x \\ 0 \\ k_t^z \end{pmatrix} \exp i(-2\omega t + \phi + 2\vartheta) \right),
\end{aligned} \tag{24}$$

where the main difference from the previous p case is that, now, K_i^H and K_t^H are real values, unlike K_t^E . Thus, the tilt angle and ellipticity of the ellipse described by the $\vec{H}$ field in the incidence plane happen to be the same as before for $\vec{E}$ in the p case (dotted line in Fig. 2), as well as the shape of the ellipse described by the tip of the Poynting vector. But the direction of the mean value of the Poynting vector now is given by

$$\eta_S = \tan^{-1} \left(\text{Re} \left(\frac{K_t^H}{|K_t^H|} k_t^x \right) / \text{Re} \left(\frac{K_t^H}{|K_t^H|} k_t^z \right) \right) = \tan^{-1} (k_t^x / k_t^z). \tag{25}$$

Note that this expression coincides with formula (25) in [17]. The dotted-dashed line ellipse in Fig. 2 illustrates the results for this s polarization. The thick line is its mean value. Explicitly, we have $\eta_S = 37.18^\circ$ and $\rho = 0.39 - 0.38i$, $\psi = 66.18^\circ$, $|\chi| = 17.94^\circ$, according to expression (38) in [17]. For the case of the second transparent medium ($n_t = 1.5$, $\kappa_t = 0.0$), we have $\eta_S = 41.04^\circ$, perpendicular to the vibration of the linear $\vec{H}$ field, as expected.

For extra information, in Fig. 2, we have represented the size of the ellipses of the transmitted Poynting vectors to be proportional to the transmitted power for polarizations s and p , assuming incident light with equal power in both polarizations. Thus, the transmitted powers can be calculated from the overall energy balance [23], giving (for $n_t = 1.5$, $\kappa_t = 1.1$):

$$\begin{aligned}
T_\parallel &= 1 - |r_\parallel|^2 = 1 - 0.31 = 0.69, \quad T_\perp = 1 - |r_\perp|^2 \\
&= 1 - 0.76 = 0.24.
\end{aligned} \tag{26}$$

These quantities have to be proportional to the projection of the mean Poynting vectors p and s , respectively, in the direction normal to the interface (i.e., the z axis).

It is interesting to note that with our formulation, we anticipate that all the ellipses calculated above and sketched in Fig. 2 (dotted line describing either the $\vec{E}$ or $\vec{H}$ field, dashed line for the Poynting vector of the p case or the dotted-dashed line corresponding to the s case) have the same value for their ellipticity $|\chi|$. Let us check this statement: (i) expressions in Eq. (17) calculate $|\chi|$ for the $\vec{E}$ field in the p case; (ii) for the Poynting vector of the p case, the expression in Eq. (20) is needed; (iii) considering the transmitted $\vec{H}$ field in the expression in Eq. (22), one may find its $|\chi|$; and (iv) for the Poynting vector of the s case, the expression in Eq. (24) is required. In all four cases, the computed $|\chi|$ value is the same.

The same formulas also imply that the minor semiaxis of the ellipse of the field (for either $\vec{E}$ or for $\vec{H}$, equivalently) does not point in the direction of the wave propagation in the second medium, as clearly illustrated in Fig. 2. As Fig. 2 also shows, the major semiaxis of the ellipse of the fields in the incidence plane and the major semiaxis of the ellipse traced by the tip of the Poynting vector are indeed always mutually perpendicular.

Of course, when k_t^z is not complex all the ellipses evolve to become the lines corresponding to their major axes, as expected for a standard transverse electromagnetic (TEM) wave. Table 1 summarizes the main numerical results for our two virtual cases.

5. CONCLUSION

We have presented a compact formulation of the problem of light reflection and refraction at a transparent-absorbing isotropic interface, including the study of the Poynting vector of transmitted light. We take advantage of the fact that the field perpendicular to the incidence plane ($\vec{H}$ for p polarization and $\vec{E}$ for s) has only one Cartesian component (in our case, y) and, subsequently, all other field components of the problem may

Table 1. Comparison of the Most Significant Results for Our Two Virtual Refraction Cases

Propagation case	$n_i = 1.0, \kappa_i = 0.0$	$n_i = 1.5, \kappa_i = 0.0$	$n_i = 1.5, \kappa_i = 1.1$	
	Incidence angle $\theta = 80^\circ$	p or s polarization	p polarization	s polarization
	Propagation angle φ	41.04°	37.18°	37.18°
	$R_k \equiv k_i^z / k_i^x$	6.52 ($\in \mathbb{R}$)	$7.48 + 7.32i$ ($\in \mathbb{C}$)	$7.48 + 7.32i$ ($\in \mathbb{C}$)
E or H	ψ (angle major axis)	−41.04° (line E, H)	−23.82° (E field)	−23.82° (H field)
	$ \chi $ (ellipticity)	0.0 (line for E or H)	17.94° (E field)	17.94° (H field)
Poynting vector	η (mean Poynting v.)	41.04°	$\eta_p = 10.47^\circ$	$\eta_s = 37.18^\circ$
	ψ (angle major axis)	41.04°	66.18°	66.18°
	$ \chi $ (ellipticity)	0.0	17.94°	17.94°

be expressed in terms of the cited perpendicular field. Overall, in comparison to the Fresnel formulas, we define and calculate a series of coefficients for the p case ($r_p^y, t_p^y, r_p^x, t_p^x, r_p^z, t_p^z$) and for the s case ($r_s^y, t_s^y, r_s^x, t_s^x, r_s^z, t_s^z$) that fully describes the ratios among all the amplitudes of all the fields. Of course, in the case of a transparent second medium, our formulation completely coincides with the Fresnel coefficients, and we have maintained the labels $\parallel, \perp$ in our expressions in Eqs. (16) and (23). But in the most general case, we have shown (completing results of several previous papers) that

1. for p polarization, the transmission coefficients for the normal and tangential $\vec{E}$ fields are different;
2. accordingly, the $\vec{E}$ field describes an ellipse in the incidence plane;
3. the direction and shape of this ellipse can be computed directly from the values of the two components of the two wave vectors k_i^x, k_i^z ;
4. the tip of the Poynting vector also describes an ellipse around its mean value in the incidence plane;
5. the direction and shape of this ellipse is again computed directly from the two components of the two wave vectors k_i^x, k_i^z ;
6. a completely equivalent behavior is demonstrated for s polarization, now for the shape and orientation of the ellipses for the $\vec{H}$ fields;
7. however, the direction of the mean value of the Poynting vector is different in the p and s cases; this direction always depends on the components of the two wave vectors k_i^x, k_i^z and additionally on the dielectric constant of the second medium (in the p case).

In summary, we have presented a compact formalism that clarifies the methodology to fully address the problem of the electromagnetic interactions at the interface between transparent and absorbing isotropic materials. Our approach is very useful since it is able to give all the relevant and detailed results (previously scattered in a set of papers) in a unified procedure for p and s cases, overall requiring reduced mathematical effort.

Regarding the experimental observation of the above results, in [8,18], two different configurations have been proposed. We are also considering the use of a thin wedge of a weakly absorbing material such as ZnSe. Clearly, any experiment is quite difficult, since the material must be absorbing, and yet the splitting of the transmitted beams must be measured. Moreover, in practice, the incident beam has to be a laser (Gaussian) beam. Thus, the above theory, valid for an incident plane wave, must be adapted (for

example, by representing a Gaussian beam as a superposition of a few plane waves).

Funding. Ministerio de Ciencia e Innovación (PID2019-104268GB-C22 / AEI/10.13039/501100011033).

Disclosures. The authors declare no conflicts of interest.

Data availability. No data were generated or analyzed in the presented research.

REFERENCES

1. M. Born and E. Wolf, *Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light*, 7th ed. (Cambridge University, 1999).
2. G. R. Fowles, *Introduction to Modern Optics*, 1975 (Holt, Rinehart and Winston, 2010), Vol. 8, pp. 33–38.
3. L. Novotny and B. Hecht, *Principles of Nano-Optics* (Cambridge University, 2012).
4. B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics*, Wiley Series in Pure and Applied Optics (Wiley, 2013).
5. E. Hecht, *Optics*, 4th ed. (2002).
6. L. Pincherle, "Refraction of plane non-uniform electromagnetic waves between absorbing media," *Phys. Rev.* **72**, 232 (1947).
7. R. J. Bell, K. R. Armstrong, C. S. Nichols, and R. W. Bradley, "Generalized laws of refraction and reflection," *J. Opt. Soc. Am.* **59**, 187–189 (1969).
8. M. A. Dupertuis, M. Proctor, and B. Acklin, "Generalization of complex Snell–Descartes and Fresnel laws," *J. Opt. Soc. Am. A* **11**, 1159–1166 (1994).
9. M. A. Dupertuis, B. Acklin, and M. Proctor, "Generalized energy balance and reciprocity relations for thin-film optics," *J. Opt. Soc. Am. A* **11**, 1167–1174 (1994).
10. C. Alberdi, S. Alfonso, M. Berroguí, J. M. Diñeiro, C. Sáenz, and B. Hernández, "Field and Poynting vectors of homogeneous waves in uniaxial and absorbing dielectric media," *J. Mod. Opt.* **49**, 1553–1566 (2002).
11. J. M. Diñeiro, M. Berroguí, S. Alfonso, C. Alberdi, B. Hernández, and C. Sáenz, "Complex unitary vectors for the direction of propagation and for the polarization of electromagnetic waves in uniaxial and absorbing dielectric media," *J. Opt. Soc. Am. A* **24**, 1767–1775 (2007).
12. S. Alfonso, C. Alberdi, J. M. Diñeiro, M. Berroguí, B. Hernandez, and C. Sáenz, "Complex unitary vectors for the direction of propagation and for the polarization of electromagnetic waves in absorbing isotropic media," *J. Opt. Soc. Am. A* **21**, 1776–1784 (2004).
13. C. Alberdi, S. Alfonso, J. M. Diñeiro, B. Hernández, and C. Sáenz, "Interface between transparent isotropic and uniaxial absorbing dielectric media: equations for ray tracing and for the direction of propagation," *J. Opt. Soc. Am. A* **25**, 2476–2488 (2008).
14. I. M. Besieris, "Comment on the 'Corrected Fresnel coefficients for lossy materials'," *IEEE Antennas Propag. Mag.* **53**(4), 161–164 (2011).
15. A. I. Mahan, "Reflection and refraction at oblique incidence on a dielectric-metallic interface as a boundary value problem in electromagnetic theory," *J. Opt. Soc. Am.* **46**, 913–926 (1956).

16. P. Halevi, "Plane electromagnetic waves in material media: are they transverse waves?" *Am. J. Phys.* **48**, 861–867 (1980).
17. P. Halevi and A. Mendoza-Hernández, "Temporal and spatial behavior of the Poynting vector in dissipative media: refraction from vacuum into a medium," *J. Opt. Soc. Am.* **71**, 1238–1242 (1981).
18. P. Halevi, "Beam splitting by a plane-parallel absorptive slab," *Opt. Lett.* **7**, 469–470 (1982).
19. J. E. Vitela, "Electromagnetic waves in dissipative media revisited," *Am. J. Phys.* **72**, 393–403 (2004).
20. O. Deparis, "Poynting vector in transfer-matrix formalism for the calculation of light absorption profile in stratified isotropic optical media," *Opt. Lett.* **36**, 3960–3962 (2011).
21. L. D. Landau and E. M. Lifshitz, *Electrodynamics of Continuous Media* (Pergamon, 1984).
22. J. M. Cabrera, F. J. López, and F. Agulló-López, *Optica Electromagnética Volumen I Fundamentos*, Óptica Electromagnética (Addison Wesley, 1998).
23. H. A. Macleod, *Thin-Film Optical Filters*, 4th ed., Series in Optics and Optoelectronics (Taylor & Francis, 2010).